Karl Friston: Derealization, Consciousness Perils

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push to not feel fear slashnot feel the ego quote unquote slash etc can be dangerous not because the lesson is false though it may be but because of the social pressure around those words in fact there's a universally frowned upon quality called being egotistical ego is embedded in it and also to be seen as retreating is the same as being seen as being defensive which is the same as being fearful which is seen as being lower on the spiritual hierarchy perhaps it's true that you should feel less disquitude about certain facts though perhaps your disquitude is justified for where you are now. And that's okay. It's 100% okay. It took me about 1 year to realize that. In fact, perhaps more than one year. It took me over one year to realize that these spiritual lessons tend to not have the physics notion of path dependence. Perhaps you shouldn't uncritically adopt an eastern practice in a western world and vice versa. Or perhaps certain truths are dependent on the path that is where you are now rather than being independent of anything. Frankly, it's embarrassing to reveal myself as so selfish and malicious and cowardly, and to do so so unguardedly to Carl Fristen and even to you, as I'm a as I'm a relatively thin-kinned person, at least currently I am. But hopefully in the end, it's a net benefit. The dangers of going on to journeys of investigations into consciousness, the self, computational reality, and even the phenomenon, as it can lead to a spiral down some viciously dark paths that are at least ostensibly irrevocable while having certainty that your newfounded conception of reality is indeed truthful. While I don't like to state personal opinions because I generally don't have many strong views and the views that I do have are fickle, changing week by week sometimes. My current conclusion is that if it's not lifeaffirming, if it's not loving, if it's not something you can tell others without others being horrified and frightened for you, then use that as a sign that what you're experiencing or what you're feeling or what you think is not truthful. Please seek help. Don't suffer alone. You're not alone. And you can verify that simply by talking to as many people as you can. If you're listening to this podcast, then it's likely because you at least have some trust in it. So trust that statement or at least test that statement out. Now on to the guest. Carl Fristen is a professor of neuroscience at University College London and is one of the most cited individuals in his field. He's also the inventor of the free energy principle and is someone who's made advances in the field of active inference that's matched almost by no one. He may be one of the most insightful and brilliant people on the planet. I've been lucky enough to speak to Carl before in a substantive 4-hour podcast linked below, which is best watched at least twice to properly understand. This episode is an outlier in the tow channel because this will serve as an introduction to that episode as many after watching that were still unclear as to what the free energy principle is. If you feel like you're lost, that's okay. Let that be a theme. Understanding is less like a light switch and more like a stove top. That is it's not like all of a sudden you have an insight and then you understand some phenomenon. You went from 0 to one. It's more like you get a gradual increase in understanding where a modicum here makes more sense and then a smidg in here. As Wheeler says and I like to quote the point isn't to drink from the fire hose but instead to get wet trusting that your conscious befulement is part of the process and unconsciously more is becoming clear. Click on the timestamp in the description if you'd like to skip this intro. My name is Kaungle. I'm a Torontonian filmmaker with a background in mathematical physics dedicated to the explication of the variegated terrain of theories of everything from a theoretical physics perspective, but as well as analyzing consciousness and seeing its potential connection to fundamental reality, whatever that is. Essentially, this channel is dedicated to exploring the unerived nature of reality, the constitutional laws that govern it, provided those laws exist at all and are even knowable to us. If you enjoy witnessing and engaging with others on the topics of psychology, consciousness, physics, etc. the channel's themes, then do consider going to the Discord and the subreddit which are linked in the description. There's also a link to the Patreon that is patreon.com/kurtjongle if you'd like to support this podcast as the patrons and the sponsors are the only reasons that I'm able to have podcasts of this quality and this depth given that I can do this now full-time thanks to both the patrons and the sponsors support. Speaking of sponsors, there are two. The first sponsor is Brilliant. During the winter break, I decided to brush up on some of the fundamentals of physics, particularly with regard to information theory, as I'd like to interview Chiara Marto on constructor theory, which is heavily based in information theory. Now, information theory is predicated on entropy. At least there's a fundamental formula for entropy. So, I ended up taking the brilliant course. I challenged myself to do one lesson per day, and I took the courses random variable distributions and knowledge/ uncertainty. What I loved is that despite knowing the formula for entropy, which is essentially hammered into you as an undergraduate, it seems like it comes down from the sky arbitrarily. And with Brilliant, for the first time, I was able to see how the formula for entropy, which you're seeing right now, is actually extremely natural. And it'd be strange to define it in any other manner. There are plenty of courses and you can even learn group theory, which is what's being referenced when you hear that the standard model is predicated on U1 cross SU2 cross SU3. Those are league groups, continuous league groups. Visit brilliant.org/ org/toe to get 20% off an annual subscription and I recommend that you don't stop before four lessons. I think you'll be greatly surprised at the ease at which you can now comprehend subjects you previously had a difficult time groing. The second sponsor is Algo. Now ALGO is an end-to-end supply chain optimization software company with software that helps business users optimize sales and operations planning to avoid stockouts, reduce return and inventory writedowns while reducing inventory investment. It's a supply chain AI that drives smart ROI headed by Amjad Hussein who's been a huge supporter of this podcast since near its inception. In fact, Amjad has his own podcast on AI and consciousness and business growth. And if you'd like to support the Toe podcast, then visit the link in the description to see Amjad's podcast because subscribing to him or at least visiting supports the toe podcast indirectly. Thank you and enjoy. Let's do an introduction once more. So, what is the free energy principle specifically? specifically its relationship to a theory of everything as that's what this channel is about, right? Um so my usual response here is do you want the the high road or the or the or the low road? Um let's take the high road. Uh given that we're talking about theories of everything. Um so the free energy principle is a principle of least action. So it's a method really. It's put most simply a method that allows you to identify the most likely dynamics, the most likely way a process will unfold. What are the what is the process that we're talking about here? Um well, we're talking about things like you and me, particles or people. Um so the first thing to note is that you can um apply standard variational principles of least action to any system specifically a random dynamical system of the kind that could be described by say a launch equation that underwrites most of the physics we know. But the special thing about the free energy principle is it tries to understand the relationship of something in relation to everything else. So you're starting from the premise that something exists and then you have to think carefully your what is a thing and how do I distinguish it from something else. So immediately you start you have to introduce a partition between the states of things um that comprise states that are internal to the thing and states that are outside the thing and thereby in think carefully about how those states are coupled. And what happens is that you have to introduce a further set of states known as blanket states that constitute a markoff blanket that uh mediate a birectional exchange between the internal states and the external states. So you've got this um notion now of taking the states of any universe, any system that you want to try and understand and partitioning it or carving it into four subsets. internal, external and then sensory and active states that constitute the blanket states where the active states mediate the effects of the internal on the external, the inside on the outside and the sensory states conversely mediate the influences of the external on the internal. So with this particular partition which you wouldn't need if you wanted to now just move on and develop um say quantum mechanics or statistical mechanics or language in classical mechanics. We're now in the special case of dealing with a system that has this partition. And in this special case, those conventional variational principles of least action translate into or um can be used to associate the paths of least action, the most likely paths that the system will take, particularly the states of the particle or person that we're talking about. um we can associate those paths of least action with paths that minimize something called variational free energy. Uh so that's basically it. And then the story is well what's what's variational free energy? Well, variational free energy uh is as with all actions in physics, it's just a um a path integral or a time integral, a sum, an accumulation of uh a functional usually known as a lronian um where in this instance the fun the functional in question, this variational free energy um can be read as exactly the same quantity that statisticians um psychologists, people trying to understand data would treat as a proxy for the evidence for some model of the world. So this is known as basin model evidence. Um statistically it's also known as a marginal likelihood. the likelihood of these sensory data the sensory the path through the sensory states under a particular model that would try to explain a hypothesis that would try to explain those data. So you've got this interpretation or certainly the fact that the functional form of this variational free energy that is being minimized by the passive least action of the autonomous the active and the internal states of something a particle or a person now looks as if it is trying to minimize a quantity which provides an approximation or a negative uh approximation to marginal likelihood or model evidencing. So in other words, it looks as if the most likely paths are trying to maximize model evidence. So one can read that in many different ways. can read that. Um and um as Jakob Howi has read it, uh a philosopher and neuroscientist who um uh has a a commitment and understanding to this kind of formulation as self-evidencing, literally interacting with your environment, actively inferring, engaging with your experienced world, your eco niche in a way that looks as if you're trying to solicit the most evidence. evidence for your model of that world. This quantity is also found in machine learning. It's known as an evidence lower bound. I should um qualify or just clarify that in machine learning this free energy is the reverse of the free energy used in physics. So in machine learning they try to maximize the negative free energy um aka an evidence lower bound or elbow. um while in physics you're you're always articulating in terms of minimizing a potential energy or minimizing in this instance some variation free energy. So you'll find that uh exactly that the same functional the same quantities the same measure of the goodness of some model of how you think your sensations your sensorium was generated um in terms of um the you know this um variational free energy that leads you to an interpretation of people particle persons inferring, accumulating evidence, making inferences about the states of the world that are generating their sensations. So that in a nutshell, or perhaps not a nutshell, is is a sort of formal explanation of the free energy principle. I did sort of slip in at the beginning very much like um things which you you may be familiar with when you were at school. uh say Hamilton's principle of um spatial reaction um it is just a method you know it allows you um to identify predict simulate engineer those trajectories those paths those dynamics those processes um that will um minimize the this this action functional or the path integral of the variation of free energy. So what that means is that if you now understand um or have at hand the functional form of your variational free energy that depends upon posterior beliefs or beliefs about the outside world under a generative model quite simply a probability density over the external causes of the consequences that are the sensory states. Um then you can use that methodology, use that principle to start to simulate to build sentient artifacts that behave in the way that we assume under the free energy principle that you and I are behaving. Let's see if I can make a nutshell of the nutshell for some people who are more familiar with physics. So let's say you have a ball. You want to know the dynamics of the ball. You can understand this with Newtonian force and that's useful for certain calculations or you can think of it as minimizing a lrangeian and then that actually provides a more useful for most cases way of calculating the trajectory of the ball. So then right there when I said ball we've identified the ball. Now here you're not just limiting yourself to balls you're saying let's say cells instead of talking about people because we can break it down simply at least for me for myself I find it much more easy to think of a cell. Okay so then we want to know what are the dynamics of the cell? How's the cell, a single cell going to act? Okay, firstly the first step would be identifying that cell. And when you mention the word marov blanket, that's a way of identifying what this object is because in order to identify what an object is, it's often useful to identify what it isn't at the same time, what it is and isn't. So that marov blanket delimmits the cell. Now, is that correct so far? Absolutely. Yeah. So, you know, I think the cell is a perfect example here. You got you've got you know a cell is just defined by the boundary which would be the surface of the cell and that would essentially be the sensory states that I was talking about before the insize the intracellular states would be the internal states and then the mill in which that cell lives which is usually in in a bunch or a family of other cells uh will constitute all the external states. So you've made the very first step just talking about a cell, a thing. You've defined it in terms of this partition into internal, external, and the uh the sensory states that bound and separate the inside from the outside. You may be asking, well, where have the active states gone? for a cell. I usually like to associate those with something called actin filaments that just lie underneath the surface, the sensory states and actively push the cell surface say or the filia um the um the the mechanisms that cause motion of the cell should it move around. So those parts those states that are responsible for cellular motility. So if we're now thinking about a moving cell or a swimming cell for example, then you need to supplement that partition with with active states. So that's absolutely the first step. Um and then you're going to go on and and now translate the ball. What's the ball in in this in this metaphor? You also mentioned the word birectional earlier. And when I was learning about this, the way that I like to understand this was that in computer science, generally computer scientists think and can only think in terms of you have an input, then there's a black box and you have an output. And that's essentially computer science in a nutshell. It's a bit more complicated than that, but it's in a nutshell. It's you have an input, you perform some calculation, and you get an output. The reason why you said birectional, at least the way that I understand it, so please correct this, is it's because of actions in that model where there's input, blackbox, output, that's generally, if we want to use the term embodied cognition, more on the cognition side. So you're just this black box where you're impressed with sensory information. The input is your senses. You're sensing all of what's happening. And then this output now is the actions that generally a computer sits there, but we're creatures in this world. At least we think we're creatures in a physical world. The black box has a model. And then one way of maximizing or showing that your model is correct is by updating your model or it's by changing what you've sensed in order to match the model. Now to change what you've sensed is to act. So it's useful to think of two black boxes. So there's one that's inputting to the other and then the output goes into the other which is the external. This is the internal. This is the cell. it senses and then it acts upon the world and then that changes the world which then changes what you sense. So is that the reason that you said birectional? Yeah. No, I think that's a really important observation. Yeah.

So you know before we were talking about these states that separate the inside from the outside and I made a big thing that they comprise sensory and active states and that there is a a two-way traffic here. And that two-way traffic instantiates exactly what you're talking about, which is a circular causality. So, you know, I use the word self-evidencing and described it as um soliciting evidence for sensory data that it provides evidence for your model of what's going on. But in soliciting, this is something that a computer would not normally do. a passive or cessile artifact like a computer, it would not be in charge of nor would it terribly worry or indeed be equipped with the capacity to select the inputs to select the data say it had to analyze. So just having a statistical package you give it some data and it produces an outcome you say a peak value or some kind of inference. Um so a lot of computing and a lot of machine learning you give it data and it provides a prediction or a classification. That's not the kind of computational process that is implied by the free energy principle. It's much more as you put it um inactive. It has you know the aspirations to provide a mathematical image of the action perception cycle where crucially you have to go and decide what data to base your inferences on by acting upon the world. So that um that sort of inactive aspect is the thing that distinguishes this kind of formalism, this calculus, this basian calculus which is quintessentially inactive. You know it's not just a um you know a good computer. It can be made a good computer just by removing active states if you want to. Um but that you know that would be a limiting and a rather um um biohysically unrealizable and unsustainable kind of artifact. Uh you know you the whole point here is that there is a reciprocal exchange with the world. So your notion of having two black boxes talking to each other I think a really nice notion. Um and indeed when applying this principle of least action, the free energy principle to interesting scenarios, um one ends up very often um putting two um sets of uh internal states each equipped with Markoff blankets talking to each other. So the active states of one blanket or one particle or one person now constitute the sensory states of the other and vice versa. So you've got now two systems trying to model each other. And if their game is to effectively um provide or solicit the most evidence for their model of the rest of the world, which is just another artifact very much like you who's doing trying to do exactly the same thing. What you you what will inevitably emerge from that joint minimization of free energy or the maximization of marginal likelihood is the kind of active engagement that renders say you and me as the two black boxes renders us mutually predictable. So what we will do is we will come to share the same generative model the same narrative so that I can predict exactly what you're going to say next given that's the kind of thing that I would say because that's the sort of world that I'm trying to model. So immediately you have this um um picture of the minimization of the joint uh variation of free energy or the um the the maximization of the joint marginal likelihood for shared models. And one gets from that some quite compelling and interesting um formulations of multi- aent games, communication, synchronization between uh con specifics um that all rest upon trying to resolve uncertainty in the service of securing actively listing asking questions of the other blackbox. you asking me questions, me asking you questions um just so I can understand what's going on out there where you are going on out there. Can any system let's say it's more than two so in the example that I gave and that we were talking about there's one cell and then the external world but obviously the world is much more complex than that and let's just imagine it's slightly more complicated. There's two cells and then an external world. Is it always the case that from the perspective of one cell, you may as well just treat the external world even though it comprises an external world plus another cell out there. Can you always treat that as its own singular black box such that we can always understand the dynamics as two black boxes or is there ever a need for a third, a fourth, a fifth? Oh, I I think I think there probably would be a need if out there if the external states um are themselves um partitioned into things. So if my ex if my universe is um composed of external states that can themselves be carved into natural kinds, you know, like other artifacts and objects and air, planets, liquids, uh natural kinds. As soon as you talk about a natural kind, you're talking about a kind of thing. As soon as there is thingness in play, there's a markoff blanket. So what that tells you is in principle it would possible to take all the states that constitute say an eco niche and this um find identify the thing of interest say say me and my internal states and and then I would um identify my blanket states and then I got the rest of the states but now I can start again. I can find another set of internal states and identify its blanket states and then take those states off the table and then start again and recursively tile all the external states. So that now I've eliminated effectively external states and replace them by the internal and the blanket states of all other things. And in on that view what happens is that because there's a um a statistical insulation or separation of internal and external states by the blanket states. That means I can never see your internal states. I don't need to know your internal states. Um because everything that is u knowable about your internal states is on the surface on your blanket states. What's so now the picture that emerges is an ensemble of things natural kinds um that are coupled to each other via their blanket states. None no individual particle or person or thing um will ever be have access to the internal states of anything else. But there will be um coupling and influences that are mediated by the blanket states, the active states um and possibly the sensory states. Mathematically, you can actually have the sensory states also influencing the outside. But for simplicity, let's let's assume that sensory states do not influence external states. Uh it is just the role of active states to actually couple back to the outside. You mentioned something extremely interesting and I want to pick up on this. You said that the sensory states can influence the external states. At least that's not how I understand it. So how can a sensory state influence an external state? So I said that um mathematically it is that is possible. Um so if you if you think about um you know from a pure from the point of view of um um somebody who's trying to understand the statistical behavior the probabilistic behavior of a set of dynamics described as differential equations or you know coupled um systems. Um then the the name of the game really is to ask your what statistical dependencies that define a Markoff blanket specifically um a Markoff blanket renders the internal states conditionally independent of the external states given the Markoff blanket. So that's their role. That's what that's what is implied by a separation of the inside from the outside that all you need to know is the blanket states because given the blanket states the inside and the outside are conditionally independent. Um so the question now is what kind of sparse coupling among all the states at hand give rise to that conditional independence that definitively specifies the markoff blanket. It so happens that all you need is the following rule. Internal states can only influence themselves and active states. external states can only influence themselves and sensory states. That's all you need. So with that um condition in place, you now immediately have a markoff blanket or a particular partition. From that follows the conditional independence and therefore the interpretation of the um autonomous dynamics which is a description of the trajectories or the paths or the processes taken by the active and the internal states of a particular of a particle. Um so that means that it is actually mathematically allowable in the sense that you don't you do not destroy that uh that markoff blanket or the statistical properties of that markoff blanket. It is allowable for the sensory states to influence uh active states. And you may be thinking well that's a bit silly. you know there's no way in which my photo receptors or my uh you know my the clear in my ear materially affect the outside and I think that's absolutely true. So for things like you and me um then you would not normally expect the sensory states to influence the external states but that's not the case for much simpler things. um when um when you look at much simpler things where most of the dynamical influences the coupling that we're talking about that you'd write down in terms of the differential equations the longan equation um most of those influences are basically short range so you know we can preclude action at a distance for example if you're a single cell you know unless you're next to me I don't really know what you're up do I'm having to exclude here all sorts of interesting things about long range electrochemical communication and electrical gradients and the like but let's just take um a bag of cells um that um that just diffuse stuff locally they just they touch each other and that's the only way that they can they can influence each other is by um being proximate in some ukidian space so everything's short range now in this situation you have a very different kind structure. So here the cell the sensory states are the cell surface and when these active states that lie underneath the cell surface c change the environment they do so by pushing the cell surface into the external mill and cause changes in the sensory states the surface states of other cells. So being a cell is probably a good example of where your sensory states which just are your surface states um actually do all the heavy lifting in terms of changing the outside the spatial uh spatial relationship with all the other cells that you're co-inhabiting a particular organ uh for example. So um you know it's a little bit counterintuitive um but I think a really interesting um sort of you know um thought experiment um and actually a practical uh consideration when it comes to looking at the different ways in which um insides coupled to outsides and the nature of interactions between the internal states of something and the external states of something. Um, and we sort of take it for granted that we can do action at a distance, you know, uh, literally I can talk to you in in Canada while you're talking while I'm in London. Um, you know, that that is that that is a a beautiful example of this action to distance. Um that means that you we have quite a special um kind of coupling between our sensory states and our active states. That is not um um a gift um of very simple um cellular like structures or automter that that can't interpret signals um or be subject to or be sensitive to um influences uh at a distance. Just so you know, or for people who are listening and if this wasn't obvious enough, the free energy principle is famous for or perhaps infamous for being somewhat inscrutable. For people who have a physics background or math background, what would be the minimum prerequisites? This is how I like to conceptualize the podcast. Before I interview someone, I think about okay, what are all the pieces of knowledge I need to know prior to interviewing them. So obvious example is let's say QFT. You need to understand classical mechanics, lawrence variance, quantum mechanics, perhaps bundle theory if you want to understand Yang Mills and so on properly. So what would be the prerequisites for someone to properly understand the free energy principle? Let's say if they have a math or physics background, it wouldn't be it would be probably much less than you'd expect to encounter um at university. I think you'd certainly need to know um sort of you know the the basics of variational calculus and it would be really nice if you understood where a lronian come came from um and uh to understand the simplicity of action you know basically you know exactly Hamilton's principle of least action and how that inherits from path integrals of lrangeians and uh how that arises from um dynamical systems theory of a very simple sort where you can write down any random dynamical system as you know a stochcastic differential equation or a lounge van equation um I think that's probably quite sufficient you know what um what you would need I think um to appreciate all the interpretational richness afforded by what happens to the paths of least action when you have a mark off blanket. You need to know a little bit of probability theory. You need to know what a conditional distribution is. You need to know uh the important you what have an intuition as to what is meant by model evidence for example or marginal likelihood. So, you know, with with I would imagine you a week's um um foraging on Wikipedia, you could probably acquire that if you know if you felt flu if you were fluent already in sort of um uh in path integral like formulations of the kind you can find on Wikipedia. So that you certainly wouldn't need know what sort of bundles were. You wouldn't need category theory. You wouldn't need to know anything about gauge theories. um you you know you wouldn't even you wouldn't even know need to know about lrange multipliers you yeah but you need to know um you know uh the basics of how to uh how how to understand um differential equations with you know with random fluctuations with a probabilistic or stochastic aspect to them and how you can understand paths of least action uh in these kinds of systems your example of Wikipedia is is on point because the way that I conceptualized this interview, this podcast, and the way that we spoke about it prior to going on air via emails is it's going to be tendike or gossamer like where you have these words imagining what you're saying is like a Wikipedia page and you say certain keywords which would be if you were a Wikipedia page underlined in blue and worthy of their own investigation in and of themselves. So, some of these I'm going to pick out and interrupt you sometimes. And I apologize for that like I did earlier in order to in order to properly set the prerequisites because we had a 4hour behemoth before and that was meant to serve as an introduction to the free energy principle. However, there were some people who were still uncertain as to exactly what the free energy principle is. So, this will serve as an introduction to that introduction. Okay. You mentioned marginal likelihood evidencing and I believe you see those two as the same and I'm unclear as to what any of those two are. So can you give an explanation what is model evidencing marginal likelihood? Ex excellent question. So this is where you you would need to be slightly fluent um in the the concepts and constructs that statisticians um would bring to the table. So um if I now um want to use probability theory um to articulate the um the the job of a statistician um trying to make sense of data. Um and just note that that sense making is basically the interpretation we're putting on these paths of least action. Um then you'd you you'd start off by by um um understanding the problem in terms of um estimating under uncertainty sort of assimilating data whilst um um quantifying and accommodating uncertainty due to random effects. So you know the random effects here can be read as um the random fluctuations or the veno fluctuations or the innovations in some stochastic process from the point of view of a physicist from the point of view of statistician um those that randomness um is inherited as random effects. Observation noise for example would be one. If you were dealing with a state space model, which is really a model of a dynamical process that a statistician might use to do data assimilation given some ev given some time series data. There may be a distinction between observation noise or um um sensor noise and the noise or the random fluctuations on the states that can't be directly observed. That would be known as state noise or system noise for example. they both you you're describing the same thing but from the point of view of statistician these these the this is becomes noise or uncertainty or random effects. So how do you deal with this? But basically um you're confronted with a problem of some data and you have a hypothesis or a model about how those data were generated by some latent states sometimes known as hidden states and hidden um here I think uh just as to try and connect it to the markoff blanket they're hidden behind the markoff blanket so the statistician can if you can think of that being an internal states They're surrounded. They're enshrouded by a Marov blanket. The data is impressing themselves on this sensory veil. There's something on the outside that are latent in the sense that they're hidden behind this uh this veil. So they only have access to the data. They don't know how uh the data were caused. So for example, they might have a hypothesis that this drug treatment materially affected um some pathophysiology that reduced the symptoms that were measurable or recordable by the doctor. But it's only the observable data that the doctor can get from the patients that the statistician has at hand. Now the statistician now has to infer was there a drug effect and what she will do she will create a a generative model that um if there was a drug effect of a certain size then I'd expect a separation of the data for example now that generative model um at the end of the day can always be written down as a joint probability density um over the causes and consequences where the causes is are the latent or the hidden states that you are trying to estimate and the consequences of the observable data that you have at hand. Given that generative model, you can now estimate um given some data the most likely hidden or latent states or causes of that data. Basically, the treatment effect that you hypothesize. So that's the posterior distribution. That's the probability of some hidden state say the external states the markoff blanket given conditioned upon the data. So that's the holy grail usually of um a statistical analysis and and you know what your most classical inference for example those based on t tests or f tests um uh right through to more sophisticated and informed uh log odd ratios such as base factors. These are just scoring the evidence for an effect of interest expressed in this latent state space um in terms of the probability of one model relative to another model. And usually in science that would be the altered hypothesis versus a null hypothesis. And then you can look at the the the ratio of the likelihood of this model or more specifically the likelihood of the data given this model divided by the likelihood of the data given another model usually the null hypothesis. So to come to our example our example of um you know the statistician trying to um infer whether the drug has had a remedial effect or not. what she would do is compute the probability of her data with an effect and then compute the probability of her data if there was no effect of the drug. So the the data sets from the treatment and non-treatment group were treated as from a single group and then she would use that to make an inference that there was indeed an effect. So the that probability the probability of the data given a particular model a specific model be it the alternate hypothesis or the null hypothesis that's the marginal likelihood. So why is it called a marginal likelihood? Well, what you've done is you've estimated given your generative model the probability of the data having marginalized or averaged away the thing that you don't know which is the effect size the treatment effect. So the probability of the data given a model is the average probability of the data given that model and the effect size averaged under your beliefs about the effect size. So it that's why it's called a marginal a marginal likelihood. Um that's a great question because um you know it actually um leads one to um the genesis of variational free energy on one reading by uh in the context of Richard Fman's pathological formulation of quantum electronamics. He he had this marginalization problem. Um so he wanted to or one could read the history as him wanting um to evaluate the marginal likelihood of um um paths of say small particles. um which would necess necessarily involve um computing or marginalizing integrating of extremely high dimensional um probability distributions which is intractable. Uh so in order to solve the marginalization or the integration problem to ensure everything sums to one effectively to get your partition function um he introduced um the variational free energy. And that was a really clever move because he converted what was an intractable and nonrealizable and incomputable marginalization uh or integration problem into a tractable optimization problem simply by introducing this variational free energy which was a bound on the marginal likelihood. So that's why it's called an elbow or an evidence lower bound simply because it now becomes a computable object or mathematical functional um that you can evaluate given some data and a generative model that is always in machine learning lower than the log of the marginal likelihood. Uh in my world, the physics world, it's the other way around. But let's stick with the machine learning world. So the negative uh variation free energy of the physicist is always smaller than the thing you want to actually evaluate. So if you just optimize by pushing up the elbow or the variation free energy until you can't make it any larger and under the assumption that the bound the tightness of the bound is high, you've now got a good approximation to what you were chasing which is the marginal likelihood. And once the statistician has that marginal likelihood, she can repeat the process for another model, say the null hypothesis with no treatment effect. And then she can take the ratio of the free energies, the variational free energies, and start to compare the models using base factors and then go back to the doctor say look um I am pretty confident that that drug really did have an effect. And I can say that because the evidence in your data suggests that the marginal likelihood of a hypothesis or a model that included an effective drug was 52.6 times greater than the marginal likelihood if I assume there had been no uh no change. So that's where the notion of a marginal likelihood comes from. Um so the marginal marginal likelihood just is the evidence for a model. just is um the um the probability of some data you in the present context you know sensory or from from the point of the free energy principle the sensory states uh or the sensory trajectory say time series um um given a particular model marginalizing away all your beliefs about the particular effect size and the parameters and all all the um quantities that were responsible for gener generating those data um um uh you know um under the structure afforded by by the model. Is that is that is that clear? I mean you know these are the fundaments of um you know machine learning, data simulation, system identification uh and and you know um you could actually argue this is one way of new nuancing the scientific um process. not quite in using the rhetoric of of um Carl Pa but certainly has this poparian aspect to it that every bit of scientific inquiry um including the way we live our lives is just about hypothesis testing. Um and we just generate alternative explanations for the evidence at hand. And then we try to evaluate the evidence for this hypothesis relative to that hypothesis. Evaluate the marginal likelihood of the data at hand for this hypothesis versus that hypothesis. This model versus that model. And then we usually commit to the model with the greatest evidence. This just is evidence-based scientific uh progress. uh and you're having committed to or selected technically known as basian model selection uh this hypothesis you then think about okay what's the next space of hypothesis or what's the next portfolio of ideas or models um that I want to now go and gather evidence for I think this analogy is quite nice I'm glad you've let me down this explanatory because because a good scientist then the good one will generate new hypotheses and then the problem contending the scientist who's been who's just chasing these marginal likelihoods or um model evidences for her models of of of her fielding uh of inquiry is to design a good experiment. So here we we we we we come back to the you know the the other side of the coin that takes us um um beyond machine learning as a black box sort of you know give it it some inputs give it an output. We now come back to the design of our experiments how we actively secure and solicit the evidence that will best disambiguate among our hypotheses. And I repeat, usually when you're writing scientific papers, it's a null hypothesis versus nton hypothesis. But in real life and in real scientific thinking, people you normally have eight hypotheses or 100 hypotheses um um and they're scoring all of them um um by evaluating the marginal likelihood of the data that they have carefully solicited by uh B optimal experimental design to maximally disambiguate in terms of the marginal likelihood of the model evidence between all of their hypotheses. So that um that sort of brings another aspect of um probability theory to the table um which would be an important compliment for somebody wanting to understand basian statistics the baze aspect of course is is that as soon as you start to talk about conditional distributions of the kind that a marginal likelihood is so it's conditional in the sense that it's a probability of some data conditioned upon on a model or given a model having marginalized away all the parameters of of of that model. Um and you're building a posterior the probability of my um hidden states the parameters of my model given the data um falls out of baze rule. So we're talking about basian inference here and just straightforward probability theory uh that can be articulated in terms in terms of baze rule. But there's another really important application of BA rule um which speaks to this sort of inactive situated uh statistician where statistician now is not simply in the service of the doctor supplying her with data and asking a a particular question. Now the statistician is in charge of designing the perfect experiment that gives her the data that allows her to make the most efficient and statistically powerful um inference about whether there was a treatment effect. And that's the um that's the problem of optimal uh experimental design. There is an answer to that. is called the principle of optim bay ba's optimal design first articulated by people like um Lindley in the in the 1950s and then rearticulated by people like David McCain in active learning um and it is if you like the other side of the basian coin when it comes to um um basian assimilation of statistical assimilation of data and inference. Um, and one could also argue it's the other side of the coin when it comes to things like basing decision theory as well. But that that's a slight distraction. Uh, I think we just stick with the big idea here that you know there are good look at it like this. If it is a case that the marginal likelihood is just the probability of some data given a model and your job is to evaluate that quantity and usually optimize the um optimize the uh the marginal likelihood or maximize the um the evidence for your model. You can either change the model or you can change the data. Um and that means that you've got sort of two uh two problems to solve as it were. If you cast this as an optimality problem as a maximization of an evidence lower bound um or a minimization of variation free energy from a from a finesesque point of view. Um and that's I think comes back to your your original uh point about the sort of two black boxes, two computers inferring each other. um you know where there are uh I can either change my mind about you uh or I can try and change you by querying you, asking a question, performing an experiment upon you to get some better um better data from you that resolves my uncertainty about you. Is there a relationship between changing your evidence versus changing your model and the psychoanalytic repression? So repression would be saying you know what that data doesn't exist let me ignore it versus the perhaps a cognitive behavioral therapy technique would be to say no go out and perhaps update your model of it view it differently view the maybe it's our arachnophobia for example I think it's a beautiful analogy and you know not probably beyond an analogy there are people in computational psychiatry who think exactly like that um and you know um you can find articles trends in cognitive science reviews of the kind you know sort that deal with you know how do you how do you resolve cognitive dissonance how do you understand wishful thinking um so examples you I think that's a a perfect example of this way of maximizing the evidence for your models of the world in that case it sounds like at least the colloquial understanding maybe in the west or maybe throughout the world I'm unsure is that you should update your model you shouldn't try to change the evidence to match your model, you should update your model. And in fact, there are some political implications. I'm sure people shout that on the street across various political divides. So, is this an argument to say that updating the model is actually the one that should be primary? Because the way that I understand from how you've explained it is that both are equally valid. However, when I look across the world and I look at how we think just in terms of common sense psychological advice, at least to us in the 21st century, it sounds like you should update your model. Um but but I think you're right to say that you know it would be um disingenuous from a mathematical point of view to privilege the model versus the data. Um um so how does one choose between those mathematically? How does one I don't I I don't think you can in the sense you know you you you can't you can't spit a coin. A coin has two two sides to it. You know, um the whole point of this inactivist um take on sensemaking um is that it, you know, you you have to optimally go and listen to the right kind of news, talk to the right kind of people um in order to solicit the right kind of data that's going to maximize the evidence for your model. Whilst at the same time, you are updating your model. um sometimes known as basing belief updating um um or certain basing model selection for example. So the two things go hand in hand. I think one way of resolving or answering your question is is just to um say in a political um um or or um um perhaps even sort of um um communication science uh sense or at least sort of the social aspects of it. Um what you're what you're saying is that um there is now once you acknowledge that you also are in charge of getting the data. Um then um then you also have to contend with the problems of how much weight you afford to that data versus that data and where you get your data from. Which news channels do you uh do you um forage? Who do you listen to? Who do you trust? Who do you ignore? Um, and we do this all the time. We do this in a very skillful way. So, personally, um, in terms of deploying, um, our attention. Um so if you're an engineer um and you were thinking about assimilating visual information using a calman filter for example then one of the most artful challenges or or artful resolutions that the calman filter brings to the table is to get the calman gain right. How much weight do you afford the prediction errors that supply the update term of a basin or a calman filter? Um, and we we do that on the fly all the time, right down to the level of um turning on and off the attention or the precision or the gain um afforded visual information. So my favorite example is um psychic eye movements. Um so just to demonstrate that you could you know one one can do this experiment at home. You can just sort of look um from one side of your visual scene to the other side and back and forth and back and forth with sacchadic eye movements uh and you will not perceive the world jumping around. However, if you reproduce the same visual impressions just by palpating your eye in the spirit of helmholds. Well, what's that palpating the eye? What does that mean? Yeah, if you Well, you can literally do this now. If you just take your finger and place it gently on the outer um aspect of your eyeball and just gently press in until you see the world shift around and move around. So you can see visual motion that is a consequence of optic flow in introduced in this instance by you pushing your retina um translating your retina with the input fixed. Um, and that is basically exactly the same from the point of view of the retina, the back of your eyeball as moving the eyeball, the back of the retina with the um with the world stationary. But notice you only saw motion of the world when you were moving your eyeball. When you move your eyes in the normal way, you don't see that motion. And the reason is that your brain ignores visual input during eye movement. And that's called psychic suppression. So I think it's a beautiful example of the fact that every moment we are selectively ignoring information that is not newsworthy. We are selecting it. We don't know we're doing it, but we're doing it all the time. Of course, you know, at a much more abstract and epistemic level, exactly the same principles apply when we choose, you know, which newspapers, which blog, uh, which Twitter accounts to follow, um, you know, which friends to talk to, which universities to go to, um, is it going to be a scholaria or a Wikipedia, you know, all of these decisions, you know, that we make in terms of securing the right kind of informative, precise data that will um, literally have a salance in the sense that it complies with the principles of optimum basian design in the sense that it will resolve the most uncertainty about what I don't know. So this is you know absolutely crucial this sort of um um being in charge of the data even before thinking about the consequences of your belief updating um you know you know getting uh just knowing that you have to listen to the right data and of course the you know the um you know from the from a sort of sociopolitical point of view what we're talking about now is how to disambiguate between fake news and true news. how to um um solicit data um in a way that precludes us being isolationist or precludes us succumbing to um cultlike um um ideologies um you know precludes fundamentalism of a dysfunctional sort or why is it dysfunctional? Well, usually because that kind of fundamentalism does not speak to and does not um minimize that joint predictability or that joint free energy that we were talking about um se several minutes ago. So what what what you will find is that um that dialectic between changing your mind on the basis of some evidence and changing the evidence by uh changing where you look for that evidence that that balance is at the heart of action and perception. And indeed you you could argue that all of action is simply in the service of gathering better data. What do you mean by better? those data that res that minimize your surprise, minimize your expected surprise or your uncertainty. Resolving uncertainty about the way you think the world works and more particularly how you work in that in in that world. This sounds like a deeper issue than just a political problem. It sounds like an issue for the process of suggesting that the conclusions that one has come to was from a dispassionate assessment of the evidence. So you hear people say this generally people who like to think of themselves as extremely rational people who've come to their conclusions because they have objectively analyzed some situation. It sounds like that is a doomed project because we're all sampling different data without realizing it. Now is that the case or no? Yes. No, I think that's absolutely right. Um and I don't think there's anything bad about it. I mean that that's just the way things are. So how does one overcome that? What what do we do? How is it this the scientific process seems to work? Um well it works because um uh the operational definition of working simply is maximizing the marginal likelihood. So but you're at liberty to actually maximize the marginal likelihood by either changing the data or changing changing the model. Um so notice that in this instance you you're not going to make any false inferences. you know, provided that all the data you solicit is always very consistent with your model, you you won't need to change your mind. Um, if you've got the wrong model, the data will provide disconfirmatory evidence and and you may well to try and get data from another source. And if you can do that and continue engaging until your model is fit for purpose, what you have basically done is find a niche for yourself where your ideology, your conceptions, your understanding of the world, the world, the way the world work is perfectly fine. So you found a cultural eco niche or possibly a physiological econ. Um if you can't, you move to another one. And so now we have a model of speciation. Um um we have a model of um shared narratives, shared uh convictions that can be communicated. to supply and teach each other. Um that um is self assembling and um um self- constructing, co-created simply because you are looking for uh at the same time you're looking for information that resolves your uncertainty about your models of how the world works. uh which makes it look as if from the outside you're basically changing the world to make your hypothesis come true and that of course is just another way of describing action. So if we come away from the sort of um the exchange of information as a scientist or somebody who reads the news or somebody who's trying to do a PhD or um um indeed just you know make sense of recent political events if we can sort of bring it back down to simple things like um moving your eyes. Yeah. um in a sense that is you changing the data to match your model. So if you uh so this falls out of the free energy principle because you know if it's the case that the autonomous paths the dynamics that we um um that um that constitute our our active engagement with the world. if they're trying to minimize free energy and we in this instance minimize free energy as say prediction error or surprise um remembering that it's the negative of the marginal likelihood. If something if the probability of these data given my understanding of my construction of the world at this point in time is very very low um then it will have a very very small marginal likelihood or have a very very high free energy and therefore it will be surprising given my model of the world. Some people like to articulate the you know the that surprise in terms of a prediction error what I predicted was going to happen um is nothing like what I actually sense the data I actually got. So imagine how um imagine now you apply that sort of concept that your the autonomous states that include the active states are just in the service of minimizing prediction error. um how we would you understand that in terms of elemental movement and behavior and one way of understanding that is that basically your models of say how you're going to move are actually realized by creating data that is consistent with your predictions. So, I have in my head a a model, a hypothesis, the notion that I'm going to raise my arm. And all that's happening in that um notion is that that's my belief about state of the world. And I'm telling my spinal cord and my musculature and my muscles um that's how I expect the world to be. And then my reflexes are driven by the prediction errors driven uh by the free energy gradients literally like the ball rolling down the hill. Um to realize to change the world so it supplies the kind of data that I predicted. Uh I just wanted to mention that um that that notion of movement as simply a realization of predicted end points, intentions to move is physiologically um fully licensed by things like the equilibrium point hypothesis in motor control and motor physiology. But it's m it goes back even further to um idea motor theory. So this was in the Victorian age how people understood uh movement that basically it was a physical man manifestation of beliefs about the way my body should be posed or what my body should be doing. So that to me that is if you like an elemental and quite fundamental example of um the importance of changing the data not the model. So if you changed your model in the context of I'm going to stand up now and you looked at the data and nothing is happening, you would immediately change your mind and say, "Oh, I'm not standing up." What would that look like? That would look like Parkinson's disease. That would be a failure to properly attenuate the evidence that I'm not moving when I strongly believe that I am about to move. And of course you can understand now the pathophysiology um the um the uh of Parkinson's disease that's due to a failure of um neuronal message passing and particularly the the weight or the gain afforded propreceptive information in that light. It's a failure of active inference. is a failure of the active states to um properly uh respond to um the evidence um that um or change the evidence at hand. So this is a lovely example of ignoring selectively ignoring just like the psychic suppression with the eyes um that enables movement. If you didn't do that, we couldn't move. So what I'm trying to say is that you know right from the the inception of these kinds of this kind of thinking in the inactive domain uh in terms of ideotive theory the uh there is a there is I think a fundamental importance of um your um of the other side of the coin from changing your mind which is basically changing the data uh upon which you make you you you um which causes you to change your mind through through sense making and without that we wouldn't have action. So you can almost say that action and perception are the two sides of the same coin. The perception is changing your mind by changing your model to maximize the marginal likelihood given a particular model. Action just is changing the data by acting upon the world to maximize the marginal likelihood basically maximize changing the data. So the probability of the data given the model is so active inference or that coupling of action perception is essentially trying to realize your beliefs about or your models of your lived world. And if you're successful in doing that, you've got the right eco niche, you will have a high marginal likelihood in from an evolutionary perspective. You will have um a um a high adaptive fitness. From a social perspective, you will be uh people will listen to you and indeed your your job in do these podcasts suggests that you you do indeed solicit that kind of information uh that causes people to, you know, to to go to your channel to attend to your stuff um simply because they they think they're going to get good stuff and they're going to get uncertainty resolving stuff. Um so this this this I think is is is is uh puts the balance back on the uh on getting the data right or the data data foraging right. Okay. Well perhaps let me undermine my own credibility here. So a few weeks ago and I think about a year ago or so we talked and I had some experience about and I was worried and you said Kurt don't be worried don't I had some experience where I thought I heard a voice. It was at a in a hypnogogic state. So, I was just about to fall asleep when many people, that shouldn't be a state that I attribute much credence to, but I remember being afraid cuz I heard my wife either say okay or just some one- syllable word. And then I was so worried and I worked myself up into anxiety thinking, do I have schizophrenia? Oh my gosh, there was no other voices since then. But I was extremely anxious. And then you told me, don't worry. And even Kurt, even if there is, there's medication and many people go through psychosis and or psychotic breaks and so on. But luckily, I didn't. Luckily, there was nothing wrong with me. And your words did calm me. So, thank you for that. However, and coincidentally, perhaps not, but coincidentally, a few weeks ago, I had a strange experience, Carl, and I wasn't sure if I was going to bring this up on the podcast for one reason, because I don't want you to think that I'm someone who's beset with psychological issues, but on the other side, I am driven by a filmmaking adage says that the more personal the pain, the more widely applicable it is. And the opposite is also false. The opposite is false. So, if you try to speak generally, you end up speaking to no one. What happened a couple weeks ago was I had this sense and it was so and even to talk about it now is a bit frightening.

I had this sense that I was like, "Oh, shoot. Solopsism is everything in my head." Just that thought. And I've because I'm on this podcast and I've entertained many ideas, it's extremely taxing to do so because I try to take different people's points of views seriously. And perhaps I shouldn't, but I feel like in order to give them a fair shake, I need to embody them in some way, shape, or form. So if they say, well, all is mind, for example, that's the idealists. I'm trying to imagine what would that be like? And if someone's a materialist, well, what does it mean that this was dead and it somehow emerged that we have consciousness? Either way, the point of that is to say, I've been feeling like for almost a year and a half now, just in a void, not sure what is true and what's not true. That's fine. I can deal with that. Except that that led me to this spiraling of such it was probably the most terrifying experience I've ever had in my life. And it took me weeks and I'm still recovering from it where I felt like shoot am I all that exists? And then just that thought alone, I'm like, I don't want to believe that that could be the case. And I remember looking at my wife and thinking, is my wife even real? How um much of an imbecile? And how ungrateful and how foolish is it? And how that that I would even think that I feel bad that I would even entertain that thought. Oh my gosh, I'm scared. I don't want to go to the hospital cuz I felt like if I was to truly accept that thought, I could I was on the brink of just losing it into an insensate spiral. I became extremely scared of myself. And I think that I'm the type of person, I am an anxious person. And I am the type of person that tends to obsess over thoughts. So then I started to obsess over this. And for the past couple weeks, I was so scared. So scared of my own mind. So almost like a hypochondric for mental disorders where I didn't want to even look up what the definition of schizophrenia or psychosis was because I was afraid that I would see the signs everywhere. For example, a few days ago or about a week ago, I remember generally we all speak to ourselves in our own heads and sometimes I use the phrase so I don't like that or I like that. I use the word I in my own internal monologue but sometimes the word yeah you can do this, you can do this. So sometimes they used I and sometimes they use you. And I imagine we all do that. It's colloquially okay to exchange those two. However, then I thought, oh shoot, what if I felt distant from my thought? What if the one that's saying you is a thought speaking to me like a schizophrenic may hear? Oh my gosh. Oh my gosh. I don't want to hear this. And I was so afraid and I felt derealized and depersonalized. when you mentioned Parkinson's earlier that the difference between the model and then the data it that didn't strike me as a a symptom of Parkinson's. The way you were describing it struck me as a symptom of derealization. Anyway, I'm bringing this all up and I actually feel good about speaking about it. It somehow feels curative to even voice these concerns. I Well, actually, I don't know. I don't I don't know why I'm I'm voicing this. Mainly to mainly Okay, how about this? I just verbally cascaded incoherently perhaps. What are your thoughts on what I just said? The reason I'm saying this one more time is that people in the comment section when they view different theories of what consciousness is, what reality is? That's the theories of everything channel in a nutshell. I see many of them in a similar place where they feel lost. For example, I've heard that some people who view Yosha box ideas that this all an illusion or this all simulated consciousness, some people become suicidal. That's actually extremely terrifying. Luckily, I'm not suicidal, though. I'm scared that I could because I've heard that other people could. I'm not sure if my problem is just obsessive thoughts or if it was indicative of something else and I was so afraid. And it's it's such a scary feeling, Carl, to be afraid of one's self, to not even be able to sleep at night because you're afraid of your own thoughts and what that may lead to. Does any of that make sense? And I know that sounds like it's completely off field. Perhaps it's not. No, no, no. I think I think it follows on very gracefully, but but you've taken it right now to um some of the most important aspects of of modeling. um which we all contend with which of course is is um I mean I mentioned before you know getting models for our um um our lived worlds but of course the most important part when we're a baby of the world is our body. So the first thing we have to do is to build a a model uh of our body and um work out you know what things that we are able to move um you know hence motor babbling and and um the rattling um probably more important than that of course is um exchanges with the mother um and developing a sense that mother is an object and crucially an object that is separate from And that's quite a skillful move and it takes a lot of mind belief updating and changing your models and building a coherent um uh explanation for all your sensations both the pitic touch or the all the and the physiological intraceptive consequences of being subtled for example and being cared for um and I'm taking this uh route to the key thing uh which is basically models of self and selfhood Um so you know if if we are um coming back to this picture of uh some um mathematical universe that I've now tiled with multiple markoff blankets encapsulating or defining lots of particles um and basically we inhabit a universe that is constituted by things like us and in initially you know mom and dad and brothers and sisters and your um peers at school. Um that means that our models of our lived or experienced world have to have models of other in it. Which means you need to be able to disambiguate between sensations caused by others and sensations that you caused as a fundamental um structural aspect of the models you're going to bring to the table. is to live in a world which is constituted by creatures like you. Um that brings with it you know uh um something quite special um which is a model that encompasses selfhood. Uh so you wouldn't need this if you um were lived in a world that did not involve an exchange with other con specifics. Um but because we do live your our world is constituted by other things like us we need to be able to contextualize and make the inference I did that or you did that or it's my turn or it's your turn or you've got that intentional stance and you know I've got this intentional stance. So just having as part of the generative model or the hypothesis you know the hypothesis that I am me and you and you is something which characterizes um one would imagine very high life forms. Now we go even higher to um basically philosophers and people like you. So if you've got if you have if you spend your life worrying about um the what is the self? Does the self even exist? Well, precisely. Yeah. Well, I mean that that's that that seems to be the the the essence of your existential angst and all its attendant um inceptive and emotional um consequences. Uh you know, do I exist? Um and I think really usefully you brought to uh the table the notions of depersonalization and derealization. If this is for physicists then can you explain to your audience what those things are because I think that it it' be really useful for people that do not know what well to be quite frank Carl I didn't want to look up this exact definition because I know that I'm a validarian and I would just obsess over myself having these issues. So I can give an explanation of my experience and then perhaps you can delineate what the psychopathology is. Okay, I'll make an aside first. Some people say that what we need to do is realize that the self is an illusion. You can hear this in the more eastern end. And I know that many people who watch this channel also feel like oh they take these people seriously like I took them seriously. And I don't know if that's exactly true. Not whether or not the truth of the self being an illusion is true but whether or not that is indeed useful for everyone. It may not be. It may be at least for me it was extremely harmful and perhaps still is and it may be dependent upon where you are in psychological development, spiritual development and sometimes these lessons that are told to us by philosophers or spiritual gurus or people seem like spiritual gurus perhaps they should be taken with such a grain of salt and I've come to the conclusion that if it's not life affirming then perhaps don't follow that perhaps use that as a clue that that's a pathological path and don't do anything that can be drastic. Don't give up. Cuz I felt like I was in a state where I could snap and feel like well this is all a dream. I remember feeling looking at my arms and feeling like am I even behind like right now I'm feeling great and I know it may not seem like it but I'm speaking with you. I feel safe. I love my wife. I've actually whatever. I'm feeling happy right now and I feel like I would a month ago which is I'm behind my eyes and this is a person speaking a week, two weeks ago, three weeks ago. I was feeling like all of that could be a model. I could emulate anyone else's head. I could be anyone else. This could all be in my head. So debilitating, destabilizing. Walking around feeling like, is this a dream? Constantly checking my fingers because that's one of the signs you can check your for lucid dreaming. How many fingers do you have? And then also feeling bad like why the heck am I thinking this is a dream? Shoot. What if it is a dream? just spiraling, spiraling, tumbling, tumbling thoughts. Okay, that's what I had slash still have to some degree, though not right now. I wouldn't say that I'm over it because it was such a recent experience. I'm not going to say that I've solved it. I feel like it's tapering off luckily. Anyway, that's what I had. I would characterize that as derealization in the sense that I didn't feel like it was real. I don't know if it's depersonalization because I don't know the difference between those two, nor dissociative identity disorder. So I don't know those the differences between those. Well, but it's it's nice you bring all those three the phenology is very very closely related. But you the way you described it I thought was almost perfect. You know this um um either um a derealization um which people normally described uh describe as they're looking um they are sensing um something that doesn't quite seem real. you know, the outside world seems like it's a movie. It's not it's not it's lost that tangibility. There's no um there's no grip that can be attained that reassures you that that stuff is actually out there. The other the converse of course is I'm um I'm no longer a person. Um and and that could I think be even more frightening. And both of these I think speak to um uh speak to um this dissociation that comes with a dissolution of selfhood or at least one's um um um construction of explanations for the self in a lived world. you used a phrase which um I I think was was quite uh pertinent here which is you know one should not pursue this kind of thinking and self-exloration and introspection unless it is um affirmatory and and and self-affirmatory and in a sense that's exactly from the sort of mathematical and dry perspective of the free energy principle that just is maximizing model evidence this is self-evidencing it is you If you can affirm through all the right kind of interactions be the emotional, social, sexual, u intellectual um that this you've got the right kind of model of your world which is constituted by people like your interviewees and your wife and all your colleagues. Um then affir affirming that model simply is securing um evidence for that good model of your rich um and incultured world that just is that that that um that sort of um affirmatory uh process. However, there are dangers that you um that lurk in the shadows which you've clearly contended with. to do that properly, to be able to secure the right kind of evidence that you are um living in a world of positive um um constructive, nourishing, enabling human beings, others, other things like you. You have to be able to communicate. To be able to communicate, you actually have to entertain the idea um that um there is there is self and other. Um and simply uh you know to engage in turn taking you need to have this um this explicit part of your generative model um that I am a self. In so doing, you will also have um contextualized as part of your generative model self in a number of different states. Uh and you need to have remember before we talking about the good scientist um having a portfolio of hypotheses. We're generating a new set of hypotheses so that we can now gather evidence of these hypotheses. So that's part of uh what you are doing. I mean you're you in in doing these podcasts you are on um a journey of um developing different hypotheses about ways of being. The the danger though when it comes to different ways of um me as a self existing is that you can sometimes um lose entertain the null hypothesis that I am not a self or I do not exist in a sentient fashion in a in the fashion of having particular kinds of qualitative experiences. This is a perfectly viable hypothesis. It can be had in a non-emotional context when you meet people like philosophers entertain things like the you know the zombie hypothesis or brain of vat thought experiments. These are all really interesting interesting instances of a very high order life form thinking about well what would it be like if I was me without mehood if I was a zombie what would it be like if it wasn't actually me talking what would it be like if there was no reality it was all just me all of these alternative ways of being that speak to selfhood or its negation or its nihilism um now become plausible hypotheses for which you have to search for evidence. And of course, if you are just doing this in your head and ruminating, introspecting, there's going to be very little evidence available to you to disaffirm your hypothesis. Perhaps I'm not me. Um, if you don't actually engage with somebody else, there is no evidence at hand for to refute their hypothesis. Uh, so on the one hand, that is a remarkable capacity. The very fact you can have this hypothesis in your head. I am not me. Uh or it is not me having these thoughts. It is not me having these quantitative experiences is quite remarkable. It's a very very high level um ability which which I would imagine where you you and I are the only kinds of um sentient artifacts in the universe that can entertain that kind of counterfactual hypothesis which is necessary to explore different models. It's necessary indeed to actually just behave and have plans. You have to have counterfactual hypothesis about the future and different ways of being. Um but of course it comes with um um it comes with a price if sometimes you um entertain a hypothesis for which there is no evidence on hand that could possibly refute it. um uh and certainly if you adopt a particular behavioral strategy or engagement with the world that precludes soliciting evidence against that hypothesis that you are say you um so examples of that would be um um things like obsessional compulsive disorder or agrophobia or um your rumination disorders um where you physically um or possibly even pro-socially prevent yourself from acquiring any further information that would allow you to change your mind. And of course, once you realize that's a possibility, you um um you can always you can also get into the negatively veanced emotional aspects of entertaining these hypotheses. So if I've got schizophrenia, that now offers a very plausible hypothesis for these dissociative experiences. Why am I having these dissociative experiences? Well, one very plausible hypothesis. I am having I have a dissociative order or I have disorganized thought disorder that characterizes schizophrenia. So, this now becomes a very plausible hypothesis. How would you go secure evidence against that? Um well, it becomes very difficult because the whole pro point of having these very high order counterfactual hypotheses or models at hand to explain your experienced world. Um is that the that that they're very that they're very plausible explanations and it's very difficult um uh if you have got schizophrenia, you're not going to be able to trust any information anyway because it could be illusion hallucinatory. So, it's one of these wonderful um self-fulfilling hypotheses that are very difficult to uh to dismantle. Um and you see the same phenomenology um in in in many instances. You mentioned behavior therapy before and in a way that doesn't have this quite the same horrible existential angst that you clearly experienced. You can actually construe um many phobias for example of the same kind in the sense that if you are very frightened of spiders, you're very unlikely to actually go and solicit the evidence that would uh that would confirm the alternate hypothesis that spiders are not frightening because you just don't go near spiders. So this is the kind of hypothesis that maintains itself because it's undermined the way that you actively solicit evidence for your beliefs. But for you, of course, you you um you know, it's not a question of being frightened of spiders. It's a much more existentially deep and worrisome notion that in fact you're not you or you're not even a self um and you're not even perceiving reality. um that these kinds of hypotheses that come along qualified by or associated with um plausible explanations, I'm going mad um um are very difficult to refute in the moment, especially if you if you ruminate on them. Um they will go away as soon as you start re-engaging with with other people. That's what I found to be the case, Carl, is that the more I talk about it, the more I'm speaking with you, the speaking with people and not being so in my head almost instantaneously, it starts to dissipate. And also, just so you know, there's a danger for the people who watch this channel that it's seen as a spiritual triumphant state to be egoless, to have ego death. So then, just so you know, when I was feeling that, I was then telling myself, but am I supposed to have ego death? Is this what ego death is like? And now I feel bad. Am I just not strong enough as a person? Which then I feel like I'm less on the social hierarchy because I'm just a coward who cannot deal with who isn't psychologically strong enough to deal with potential ego death. If that was even ego death. So that those are there's so many dangers in this and I didn't realize this. I had this insight in that episode or experience that Kurt Curt Kurt something was like I was talking to myself or something was talking to me and telling me Kurt you've been thinking way too much. This is you're extremely analytical. I never would have thought of myself as someone who's too analytical before. It was clear to me that I was too analytical. And there's nothing wrong with being analytical. There's nothing wrong with thinking. But you've been thinking a bit too much. Feel more. Just ground yourself here. Engage in this world more. And and there was also this feeling that I don't love myself, which is a strange feeling, Carl, that I don't have that I'm a bit hard on myself. And I would have never said that before. I would have thought I'm not hard enough. I need to study harder. I need to prepare for these podcast harder. I need to release more. I need to learn more. Push, push, push, push. It was only then that I felt like Kurt that's been dangerous, man. Like just relax and and don't worry too much and feeling deep talking about this somewhat publicly is a bit embarrassing because I do feel like by saying this like there's still that part of me that feels like am I saying this to give my way out because I'm truly lazy and I want to say this. So I get the sympathy of for example you maybe the audience to say Kurt no no no you're doing great. So then I can get I can feel there's so much self-analyzing that's going on right here Carl. Anyway I felt like Kurt you were thinking too much. Just relax perhaps. Don't engage as deeply as you have been with these ideas. Don't entertain them as much. You maybe you're in a fragile state right now. You could do that later. Right now take it back. That's what came over me. And also this lesson that I don't have enough self-love or self-acceptance. Last night I looked up I was using a rubber band for my thoughts like self-administered aversive conditioning when you snap yourself if you're performing a bad habit. So I thought maybe I could do that with my thoughts. So anytime that I feel like that that would encourage my mind to not feel like that. I looked it up and it said that works for physical habits but not for mental habits. So stop that. And what works is ACT I believe acceptance and commitment therapy which is a CBT technique. So then last night I was telling myself, you know what? Try that out. Accept these thoughts. Now I'm not exactly sure what it means to accept them. I don't precisely think it means entertain them, but at least meet them not with horror and dread and avoidance, but just say that's okay, man. You think like that. Some people feel like that sometimes. So I started doing that. And last night was the first time in the 3 weeks since I've had this that I felt absolutely like my regular self. Almost from 5 minutes of this quote unquote acceptance therapy. And then I woke up this morning felt like my regular self. Like I'm behind my eyes. I have a I'm in this world. That was fascinating. Okay. So I said quite a few statements right there. I'm sure there's plenty you want to comment on. What I'm also interested in for you to put a pin in your hat is what the heck does this self-acceptance, self-love, if that's even the same, mean in terms of the free energy principle. Okay, so I've basically just poured myself out there. There you go. Sorry, Carl, for you to pick up all those pieces. I know it's a huge responsibility. I trust you. You're someone who studies this. You're extremely extremely bright and you have such a wealth of knowledge and I do feel like what I'm going through is not something terribly unique. I do feel like this is something that many people who are on this journey of understanding the world and consciousness and their place go through this perhaps at some point. And I think I was trying to emphasize that by saying that you know it is a gift that you're able to have these existential crises. Um you know there are dangers that lurk which you've clearly encountered but at the end of the day the very fact that you can entertain these counterfactuals is you know quite remarkable. You know, to my mind, it's it would be the highest expression of the human condition just to consider we those alternative hypotheses where bits of our humanity are just not there anymore or have a a different disposition or relationship to reality or indeed other people. There are lots of things you said made made a lot of sense from simple things like you know soliciting reassurance from others that you don't have to do um any more work in terms of preparing yourself and being a good a good u expounder of ideas or or or interviewer or bookw writer or or or whatever. Uh you know that's exactly what you should be doing. That's that's exactly securing evidence for yourself models of an affirmatory sort. Uh and the more you do that then the more uh the more that model will be fit for purpose and you will have the the right kind of marginal maximizing your marginal likelihood. But you're in a difficult position because your job is actually to explore other ways of making sense of things and other ways of thinking things. So you're you're compelled to explore alternative hypotheses. Um and you know you you've described um the you know the potential um nihilism and um um that I presume quite nightmarish um I can't now remember night nightmares but I'm I imagine that all of us do go through this when we have nightmares as children. I think you stop having true nightmares in your 30s and 40s. But uh you know the that sort of you know before you've really got that grip and maintain that grip on selfhood and you in your in your lived uh in your lived world as part of that and engage with that world before you've got there. you know, it must be absolutely awful, you know, nihilistic to to occasionally lose that and then, you know, worrying that you're never going to get it back. Does remind me of certain um certain um drug induced states. You know, I I can imagine that you're the kind of person who will be very afraid of having a very bad trip. Um and I think very frightened of having one. Yeah. Yes. That that is correct. And luckily I wasn't well what I imagined what I was going through would feels like how someone would feel on a on a terrible psychedelic trip. Yes. Um and it's interesting you bring up psychedelics because of course you know um there is a move um in sort of pharmacologically assisted talking therapies you know to to to use psychedelics simply um not to aspire to u I I don't know what ego death means but I I can have a guess at it from the point of view of meditation and mindfulness um but there's a really interesting connection between um the use of psyched elics um and the aspiration of many um meditationike practices um that would I think subsume the intent internal attention states um of um of meditation and your current uh practices of mindfulness which is really to try and redirect your attention to the sensorium and usually the interceptive part so it's it's like breathing for example so this is the exact opposite of what you've been It's exactly deploying that attention all that um um game control we're talking about in terms of selecting what kind of information sensory information sensory states to um engage and um determine your belief updating. So putting it putting all that attention out to the sensory side of your deep hierarchal models, your your constructs. Uh what you've doing is the opposite. You've you you've been actually wandering around at the deeper the highest levels of these hierarchal models that Sorry. So in back when we were initially talking I said there's a black box the input output and then there are two black boxes. It's as if you're saying pay attention to the sensory. Forget about the action. We haven't talked much about the action, though obviously as in body creatures, sensory and action are tied. It's as if what I've been doing was staying within the box and creating my little sensations and actions within there as little loops. And you're saying pay attention to the senses that come from the outside. Is that what you're saying? Or Yes. Well, I'm not saying you should do you you will do um everything uh in the right way under the free energy principle. What I'm saying is that the what I'm just saying is that the the skilled practitioners of mindfulness and meditation I would imagine of the kind that would lead to ego death or I'm not quite sure what that means. Um it um uh w would employ exactly the same kind of internally mediated and sometimes if you're very skilled valitionally called forth mechanisms um that are actually um targeted by psychedelics there that require you now to do all your belief updating and evidence assimilation and sense at the much more the input level of of the black box not at the if you like the highest level which you could think of as an output in terms of selecting the plans or what am I going to do next and what kind of person am I narratives that we give ourselves that contextualize the way that we behave and the other things that we do um so you know I think there mechanistically a really interesting connection between the notion of you ruminating going in circles around your head exploring every darker and darker um uh um corners of your mind and different hypotheses for which there's very little evidence available simply because you're not attending to the sensorium. You're not talking to anybody. Yeah. You're not for example, you're not you're not focusing on your breathing and indeed you try and supplement that with an elastic band just to get yourself back to the sensor. That's interesting. certainly. So, um I I think that you know you might you may well guess you probably are um skilled in meditation. Um but that kind I think the objective of um becoming a skilled practitioner of mindfulness or meditation is simply to get some valitional control over that sort of attention um that is paid to these deeper machinations versus attention that is paid to the sensorium that underwrites the psychedelic aspect. the um the allure of just sensory uh patterns and um um the sensations um and textures that you get, you know, when when taking psychedelics. So, you know, getting control of the balance um um so it's not the fact you're attending to your breathing which is important. It's the fact that you can valitionally redirect your attention away from selfhood control of it. So, but you seem to have got control um by this acceptance. And it strikes me that, you know, in the absence of um feelings of an emotional sort that were articulated in your description in terms of, you know, I'm an awful person. I'm not fit for purpose. You know, I'm not well prepared. Um I um I'm not able to man up and own this ego uh ego death. uh all of these sort of personal but quite emotional constructs effective veilanced um aspects they are also hypotheses they're also explanations so the idea is that you know anything that you can talk about has to be part of an explanation for particular ways of being that you know I can be a worthless person or in this sense I am a worthless person I'm not a worthless person uh you know and you since I I'm in a state of anxiety I'm not in a state of anxiety we have to recognize when we are anxious. Anxious is just a state of being um which is necessarily called for in certain situations and rebalance this attentional precision waiting in um in computational psychiatry. Um to make your belief updating fit for purpose in this partic in this particular context. Um, what I'm trying to get to is why acceptance might have worked. Because I imagine that if you were suddenly found yourself ruminating and locked into those ruminations, um, then very much akin to someone on a bad trip who thinks, "Ah, I am going to be locked in this forever. This is going to be my uncertain state of being for eternity because there is no reality today." That's how it feels. Very how felt. Yeah. that um that feeling will well if that is true I must be feeling anxious. You are then going to look for evidence that you're feeling anxious. uh um and because you are anxious uh you will experience certain cardiac acceleration certain uh intraceptive um flight or fright like responses that will supply evidence that you are anxious and then that becomes evidence yes I am right I am now in a state of nihilism I do not exist and I should be anxious about that and yes I am anxious so you get into a um a vicious circle. So this is a sort of uh good oldfashioned cognitive behavioral explanation for things like panic attacks that you know uh you quite reasonably in in an entirely basoptal way make sense of intraceptive bodily gut feelings literally um you make sense of them with the hypothesis ah I must be in an anxious state of being and that conclusion that hypothesis then generates autonomic actions that are realized reflexively in the way we talked about um when we're looking at the Parkinson's idea um and that will be reflected in terms of cardiac acceleration uh neuroendocrine releases uh into your body and your body will change and your body will supply signals interceptive evidence yes I am anxious so it's again a self-fulfilling prophecy in exactly the same way the idea theory means that raising my hand is just uh a realization of my beliefs but I can literally raise my levels of anxiety in exactly the same way but invisibly from the outside but not when I can sense my own body. So if you got yourself caught up into this joint hypothesis, I am in a nihilistic existential state and I should be really worried about this and anxious about this. Then it is quite understandable that you are going to use signals of anxiety and angst generated by your body as evidence that yes, I am now um divorced from reality. I'm an person. I am, you know, in a in a dark and nihilistic um um place and there could be no other place. There could be no reality. That's a hypothesis. If you can just wait for the evidence that that hypothesis is incorrect simply by letting your body calm down so there is no further evidence of any anxiety. Then you can find secure even from your own body, even without talking to somebody. You can secure evidence. Ah, the hypothesis that I am in a state of internal nilism and anxiety is a silly hypothesis because the evidence refutes it. I can now feel my you not possibly personally but you you will synthesize all your interceptive feelings um in a way that um provides evidence against the fact that now I am a deeply nihilistic um anxious state. I think that just is the motivation for the acceptance. It is pushing through to get to a state of mind and a state of body literally you know realizing your good and making sense of your gut feelings so that you can now refute the joint hypothesis that I am uh in a state of nihilism and I should be jolly frightened about this because there's no evidence that you're frightened anymore. So this was just a piece of um um hypothesis building of you know a very sophisticated and um uh uh and philosophical sort you know the highest level of exploring alternative ways of relating to the world. you now have I think a very useful insight um into what into the gift of retaining selfhood that we I think we all we all take it for granted but it's a quite a fragile thing you know uh luckily um you know most of us get through the day if not hopefully uh most of our lives you know by retaining that grip and just you know continue securing evidence that this is the right hypothesis I'm a person I am me I am functional. This is, you know, this is the way I'm meant to be and these are the kinds of things I do and the kinds of people I talk to. But that is such a fragile self assembled and has to be maintained hypothesis that you don't that you that fragility I think only is revealed occasionally and only to some people when they have the alternative hypothesis that you know I I'm not me or you know me is not quite as functional as I thought I was or I'm several me or me uh um me uh I am only me and there is no reality out there. How is it that you prevent yourself from getting into existential crisis when it seems at least to me from the outside that you study similar not theories per se but the free energy principle is about existence and is about selfhood. So, is your model so strong, Carl? Is your model already so strong that you can entertain these without worry that you'll break yours or without that being a potential possibility? Or is it because you purposefully analyze it dispassionately, analytically, intellectually, so that you're not identifying with what you are writing down on a piece of paper? How is it that you prevent yourself? Or is it constitutional, like by predelection, by personality? You're not neurotic. You're not as neurotic as I am. Um, I I think there's there's probably truth to all that. And um and by being neurotic um you know that's an entirely your neurosis are absolutely essential to get through life. Um you know I have a particular set of of neurosis um which you know I treasure and try to and try to I in the big five model. Yes. Um um but I think you're also right that um some people might call this mentalizing. I think what you were doing is that you were um in a raw enthusiastic and creative way were just exploring different hypotheses about the way that you yourself should it exist um um relates to um you the rest of your um mind and and the outside world. But for you it got conflated with um uh emotional states like anxiety um um and and your self-worth and and those evaluative uh states of being. Um so if you want to in your words be more dispassionate about this kind of mental exploration on the inside then you have to um you you have to train yourself not to conflate the two. One way of doing that is just to become to mentalize. So the mentalizing is usually taken as a way of if you like um disentangling what we what we would normally quite functionally put together certain states of being um of a sort of um intellectual sort or indeed a physical sort uh or just a contextual sort usually come along with some uh emotional veilanced effective um um uh aspect simply because that's the way the world is. So if you're say um um very late at night um you're normally going to feel fatigued and slight you know uh slightly less robust you know in in a way uh in physically. Now it doesn't have to be like that but that's the way that normally the world works and that's the way that you've learned to model yourself. Um, if you train to be a professional, say uh like an airline pilot trying to land a plane at night or like a psychiatrist who has to deal with these um has to see people going through um these um um emotional or veanced or effective um episodes. Whether it's a sort of um affected in the sense of uh your mania or depression or whether it's um the the you know the the comorbidity associated with psychotic states uh say schizophffected schizophrenia sorry schizopffective disorders. What's that? It's a mixture of um um abnormal mood um with having psychosis of the kind you'd associate with schizophrenia. Um so it's having an emotional aspect to say delusions and hallucinations for example. Oh I thought that those so those don't go hand in hand. So some people have delusions but they're okay with it. Yes. Yes. Um, some people have hallucinations and and would not uh would not even seek med medical help or their family and friends would not not seek medical help. Yeah. So there are people out there. I guess my point you know um here is that you can mentalize it and build professional defenses against the emotional aspects and I think that's what I've done. So I think about these things as a doctor or a mathematician not as a person. And if you want uh if you want an honest answer, the other trick uh is to is is to uh not use psychedelics, but you should you should try smoking. So smoking is is smoking and cups of tea are great. Smoking what? Cigarettes or I actually smoke a pipe but cigarettes will do in an emergency. So there are certain neurot manipulations and neurotransmitters. It's the first time a doctor has recommended smoking. Yes. Well, I hope we have a select audience for this. Well, yeah, continue. Could you finish up your thought? I'm sorry. No, no, no, no, no. You were saying the smoking of the pipe does what? Because I'm not the type of I don't smoke and nor do I want to smoke, but you were saying that the smoking does what was the purpose of the smoking? It's just a way of using um nuancing your neurotransmitters that control that synaptic gain or the gain and the weight you for different uh different parts of your um belief updating. So it has the opposite effect of psychedelics basically. So that's why people like tea and you know have a cup of tea calm down you know stay calm and and carry on or what is wherever it is nice cup of tea cigarette and cup of coffee that you know that that'll sort you out and it's because all of these um drugs the caffeine the nicotine they all act on the same kind of um receptors um at different levels in your neuronal hierarchy um that are responsible for setting that gain control and gating the attentional and the the attenuation aspects that we're talking about. So, it's just a way of getting physical control over your hierarchal belief updating. A bit like you're about I would have thought that the caffeine would have the opposite effect of increasing your anxiety. Not not not necessarily. No. Um you know, if well, it clearly depends. um if you're very agitated um um then um that could certainly be the case. But if you're if you are worried that you might be anxious and in fact you're not then desensitizing certain um certain interceptive sensations um that register the fact in fact you're not terribly your heart isn't pounding that actually calm you down a little bit. Uh if it goes too far then beta blockers will be another nice example of that's what I used actually. Oh right. Oh well there you are. So what I'm trying to say is sometimes the with skillful use the use of a tobacco pipe can have the same kind of effects as a beta blocker. That's all I would say. All right. All right. Professor again thank you much much safer to stick with beta blockers and you won't get mouth cancer or anything. So, so if you could I found the beta blockers work and there's also some evidence of fear extinction. Have you heard of that? There's this professor named Mel Kend who I'm going to speak to at some point. She takes people through phobias where she gets them to be extremely frightened. She actually makes them almost manic or panicking at least and then she gives them a beta blocker and then there they still feel fear that day. It's only the next day when there's memory reconsolidation and the emotional tags are removed because of the beta blocker. So you have to have a good night's sleep and then the fear seems to dissipate drastically for incurable PTSD. I was trying to do that to myself. So when I was feeling some fear, I would take a beta blocker and then I would feel the fear less. Still wouldn't be zero. But then thinking, okay, great. Maybe this will be like my own intervention of a fear extinction technique. Yes, I I was that's very clever idea actually to combine beta blockers with what extinction or some people call flooding. So that yeah so that that's exactly the kind of way I was um counseling you you would use people who smoke use smoking to moderate in the right some tea yeah there you have it okay that's very interesting thank you so much I we'll talk again and perhaps the next time we speak there were some questions that I didn't even get to from Michael Leaven from Professor Norman Wildberger who's a mathematician and and there there's quite a few Carl, I kept you on for so long and this has gone well, this didn't I didn't intend for it to go in this direction, but I'm extremely glad it has. And maybe my episode can serve and your advice episode plus the advice can serve as a cautionary tale and as well as some practical advice as to what to do, how to avoid situations like this when exploring topics like this. Thank you so much, professor. I appreciate it. My pleasure. I look forward to the next time. The podcast is now finished. If you'd like to support conversations like this, then do consider going to patreon.com/cur tj a i m u n g- a l. That is kurtjong. It's support from the patrons and from the sponsors that allow me to do this full-time. Every dollar helps tremendously. Thank you.